Tutorial: Reverse Engineering, Part 1

Ibrahim Konare, Ryan Evans, Boyang Wang

Data and System Security Lab (DaSec), Department of ECE, University of Cincinnati
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1 Introduction

What This Document is About. This document introduces the basics of software reverse engineering using Ghidra, a free open-source tool that was developed by the NSA. In this tutorial (Part 1 and Part 2), we will walk through the installation and setup of Ghidra (Part 1), explain its main interface features (Part 2), and go over some foundational reverse engineering techniques (Part 2), with hands-on examples to get started. As we go through the examples, we will provide a workflow that will be beneficial to replicate and understand how secure coding practices impacts software development in various engineering domains. Readers, who have VM, Linux, and Ghidra installed already, can skip the details of Part 1 and proceed to Part 2 directly.

Pre-Requisites. To follow the content of this document, readers are expected to have a basic understanding of C or C++, as well as some background with computer architecture fundamentals, such as instruction sets and assembly language.

Note. Reverse engineering is not limited to Ghidra alone. Other tools, such as IDA Pro and Binary Nijia, are also available. This tutorial aims to provide introductory level of knowledge and basic understanding on (software) reverse engineering before moving on to more advanced reverse engineering tasks.

GitHub Repository. Our tutorial document (Part 1 and Part 2) as well as the code and tutorial videos can be found on our GitHub repository[1].

[1] https://github.com/UCdasec/Tutorial-Ghidra

2 Software and Hardware Settings

Before using Ghidra, it's important to consider system requirements. Since Ghidra is often used for reverse engineering, especially malware analysis, certain precautions are necessary. In our case, we'll use a virtual machine (e.g. Ubuntu) to create an isolate environment, keeping our local system safe. The following are hardware and software specifications of the system we used throughout this guide and recommendations to adhere to other applicable systems.

- Hardware: For our designated station, we are using a desktop with an AMD RYZEN 5900 CPU, 16 GB memory, and a RTX 3060 NVidia GPU. We recommended the following hardware for your host.
  - Host OS
    - Any latest Intel/Arm Processor
    - Windows 10 or later (64-bit)
    - MacOS 10.13 or later (ARM64)
    - Any modern distributions of Linux (64-bit)
  - Hardware
    - 16gb+ of RAM
    - 500gb+ of disk space
    - Any mid-high end graphics card
- Software: Virtual Box and Ghidra. We use Virtual Box to run our Linux VM (Ubuntu) on our host machine. Installation instructions are described in this tutorial.

3 Installation Guide

3.1 Installation of Virtual Box

In this section, we will demonstrate how to install Virtual Box. Furthermore, this setup will allow us to run our virtual machine (e.g Ubuntu). As previously said, we will be using our dedicated host (e.g. Windows 11, 64 bit) for the duration of this tutorial. To get started, go to the official download page **https://www.virtualbox.org/wiki/Downloads** and click on **Windows hosts** to begin the download.

Once the installation has completed, run the installer and follow the steps in the wizard to complete the set up for Virtual Box.

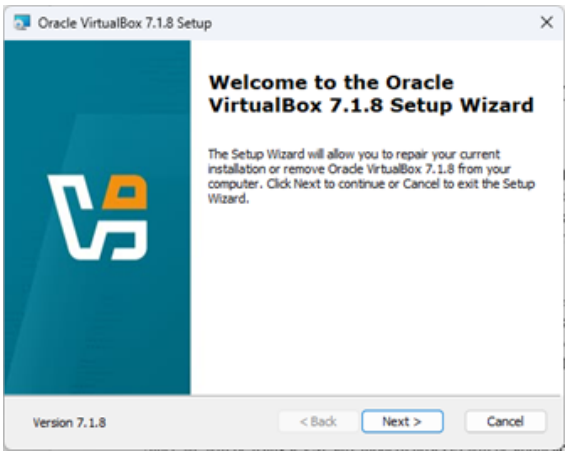

Fig. 1: Start menu of installation wizard

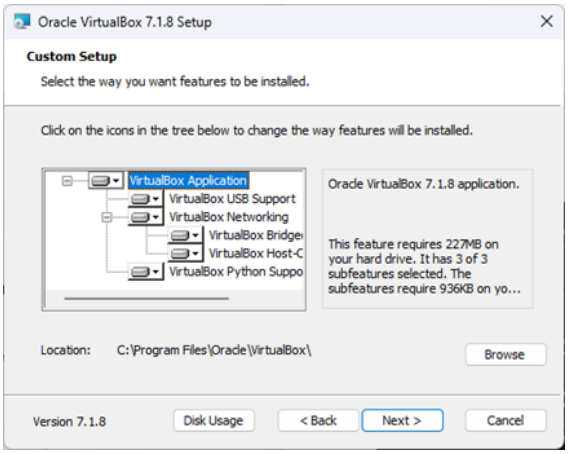

Fig. 2: Confirmation prompt to proceed with installation

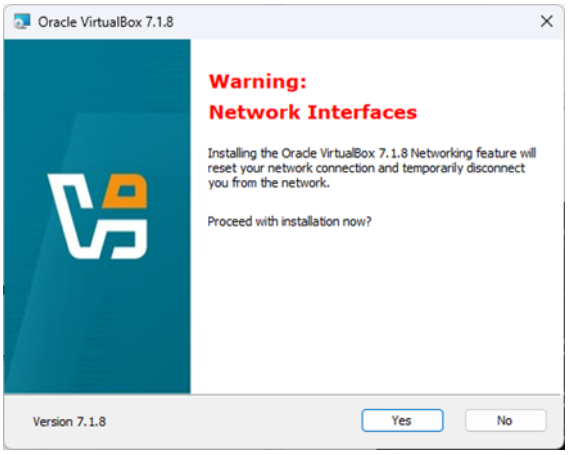

Fig. 3: Accept license agreement to continue

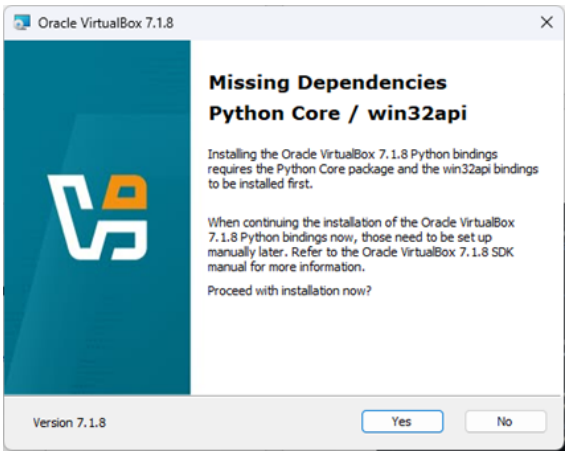

Fig. 4: Prompt to install additional dependencies

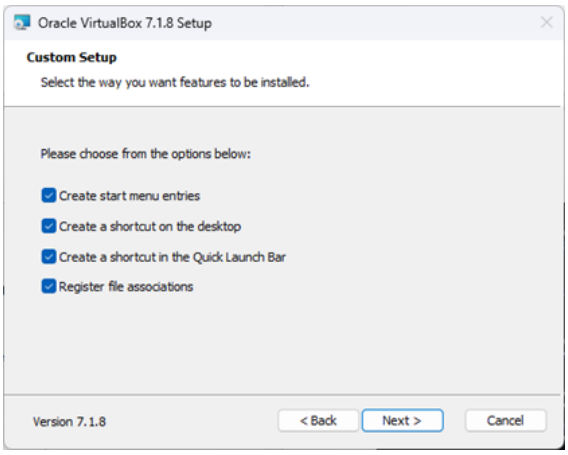

Fig. 5: Feature selection for installation

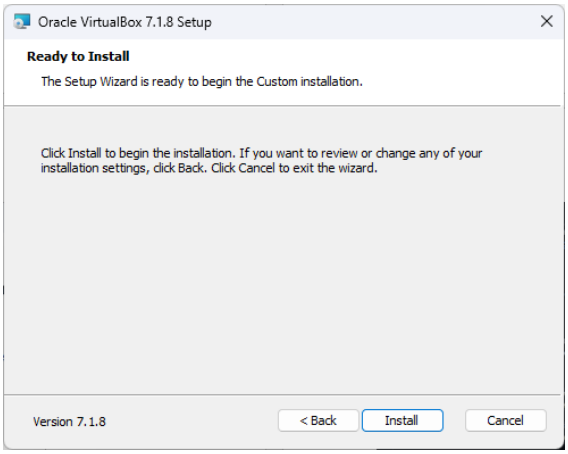

Fig. 6: Final screen of installation wizard

3.2 Installation of Ubuntu Linux

After completing the installation of the hypervisor (Virtual Box), the next step is to install a virtual machine image (ISO). In our case, we will be using Ubuntu latest version as of June 2025, which is 25.04. **https://ubuntu.com/download/desktop**

On the Ubuntu download site, there are various options for installation. Choose the one labeled as **Intel or AMD 64-bit architecture**, which is compatible to most modern systems. This is the ISO file we will be using to set up our virtual machine.

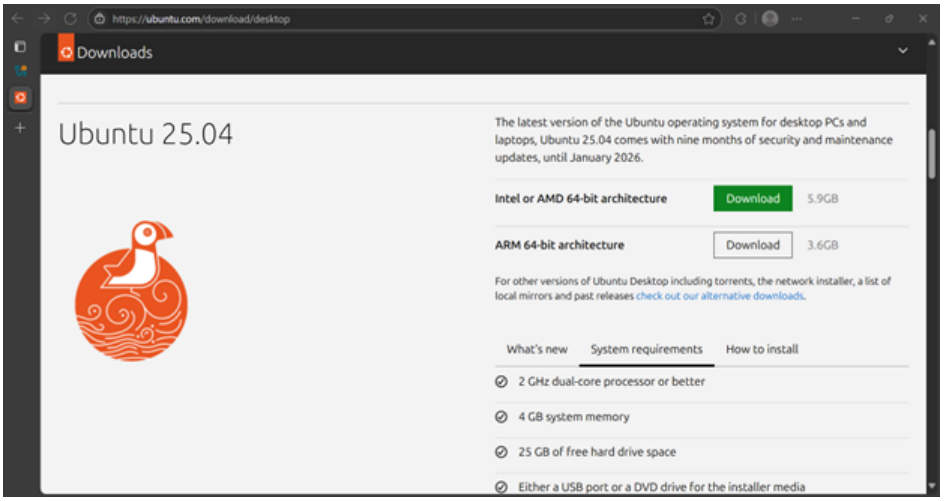

Fig. 7: Official Ubuntu download page

Once the ISO has finished downloading, we will create a new virtual machine (VM). Open VirtualBox, click on **New**, and configure the parameters as show in Figure 9. Assign a name to the virtual machine and upload the ISO file.

Important: Make sure to check the box labeled as **Skip Unattended Installation**. This helps us by retaining full control over our Ubuntu installation process.

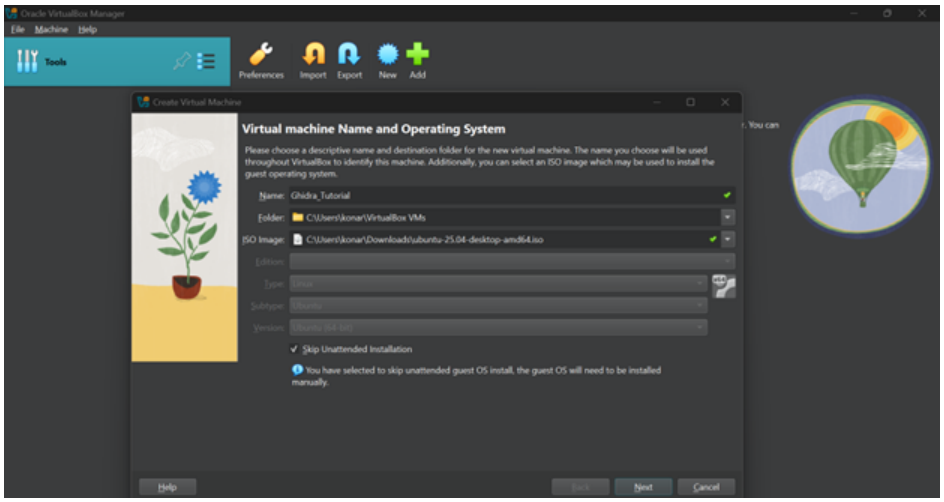

Fig. 8: Virtual Box "Create a Virtual Machine" Window

Hardware Specifications. Like any computer system, the more memory and processing power it has, the smoother the experience and workflow can execute quicker. To ensure compatibility and responsiveness using our virtual machine, we will allocate 6 GB of RAM and 4 CPUs to the VM. Additionally we will select **Enable EFI**, which gives us a way to troubleshoot any boot-related problems.

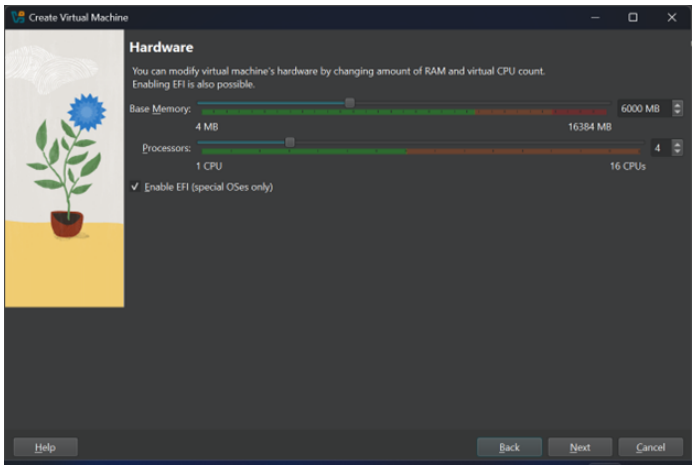

Fig. 9: Hardware Customization for VM

Storage. Apart from the Hardware Specifications, we need to consider how much storage we will allocate to the VM. If we were to analyze a large binary, we would obviously want to have an abundant amount of storage. In our case, 40 GB will be good to start with. Additional space can be added later if needed.

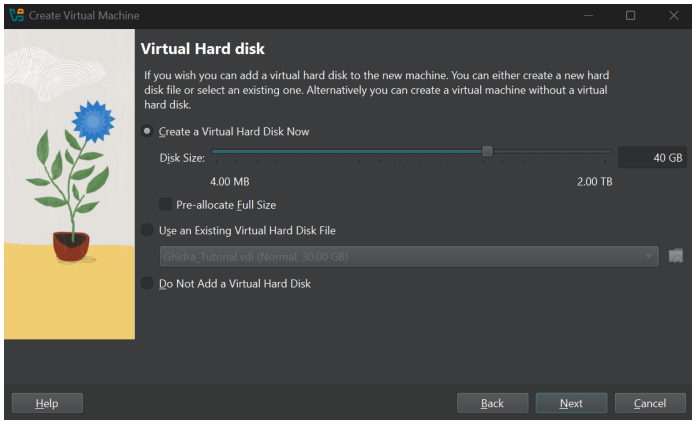

Fig. 10: Hard Disk Customization for VM

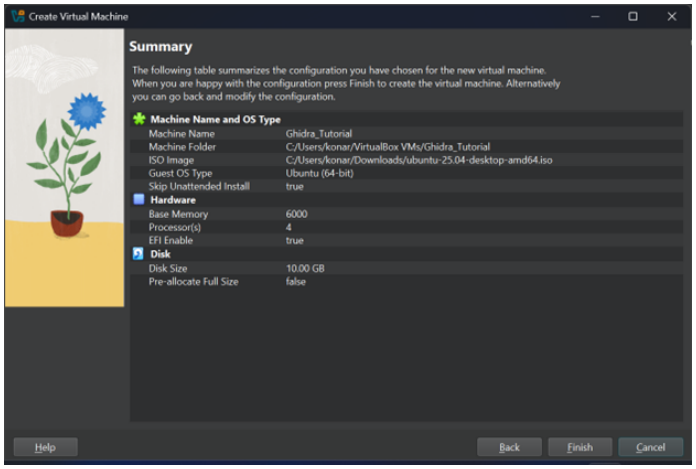

Fig. 11: Summary Part of Setup. The end of setup process for Virutal Machine

4 Ubuntu Environment Setup and Ghidra Installation

4.1 Ubuntu Enviroment Setup

After booting up the virtual machine, the Ubuntu installer will launch automatically. Follow the instructions below for the recommended installation that will be tailored for our reverse engineering task.

– Select **English** or your preferred language, then click **Next**.

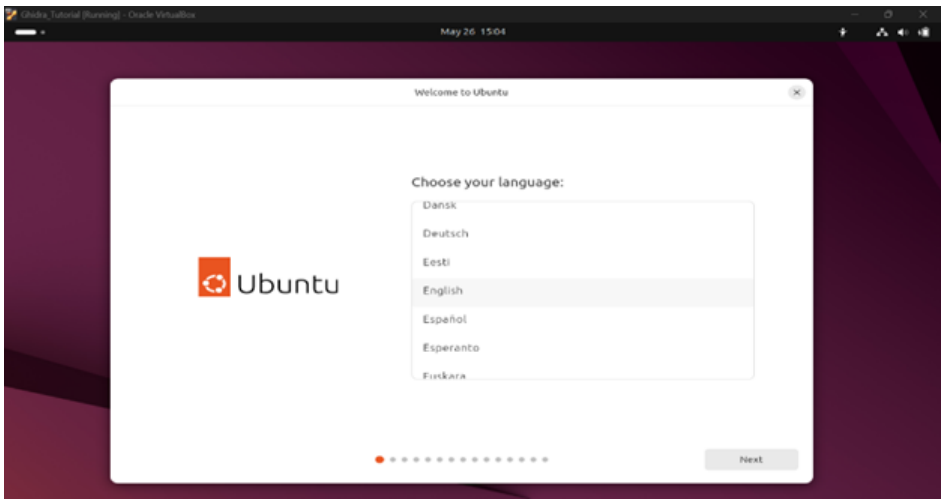

Fig. 12: Welcome to Ubuntu

– Choose your keyboard layout. If unsure, leave the default (usually **English (US)**) and click **Next**.

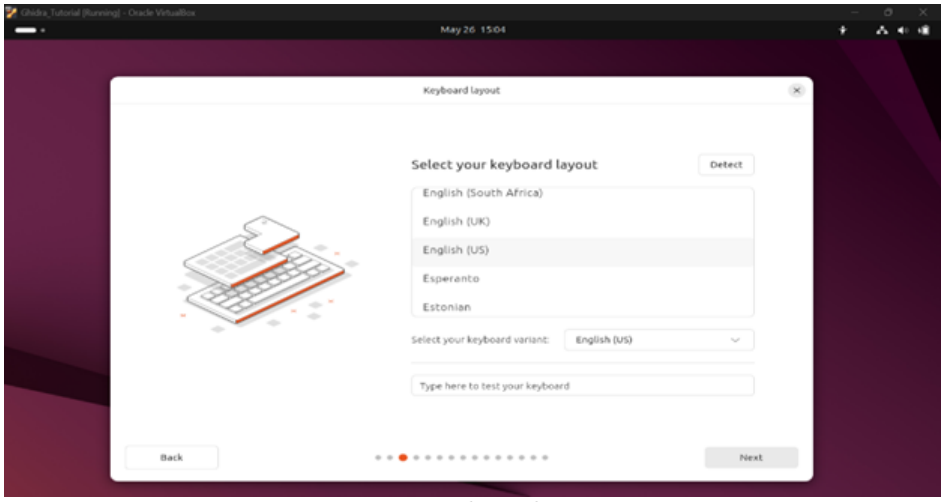

Fig. 13: Keyboard Layout

– Keep the wired connection enabled. The virtual machine uses the host's network adapter for internet access.

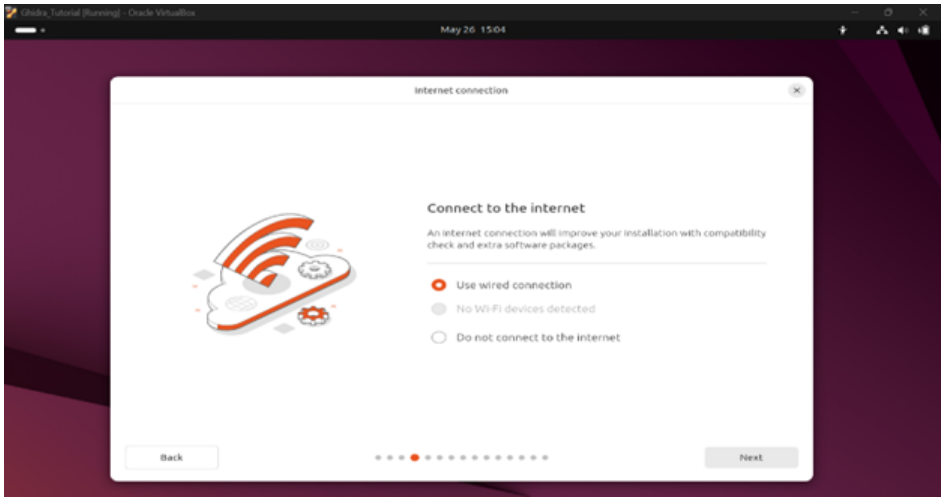

Fig. 14: Internet Connection

– Select **Install Ubuntu** to begin the full installation process.

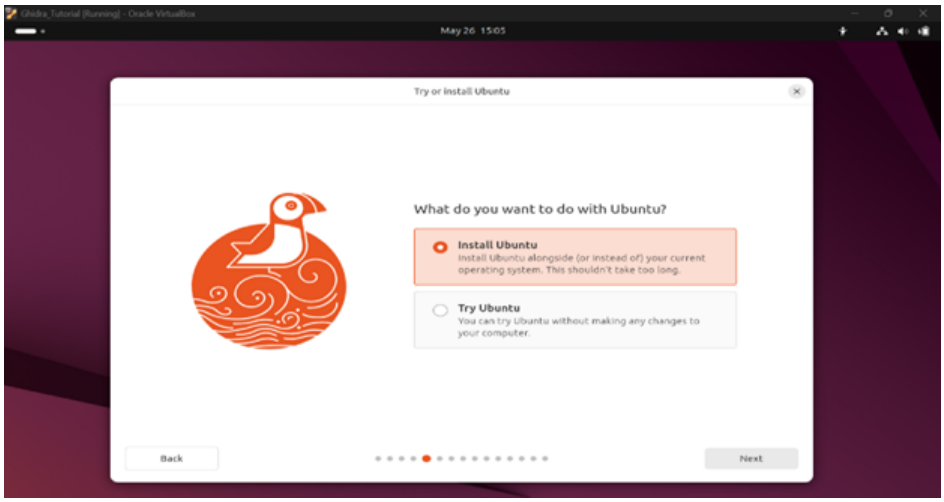

Fig. 15: Try or Install Ubuntu

– Select **Interactive Installation** (this gives more control over installed features) and click **Next**.

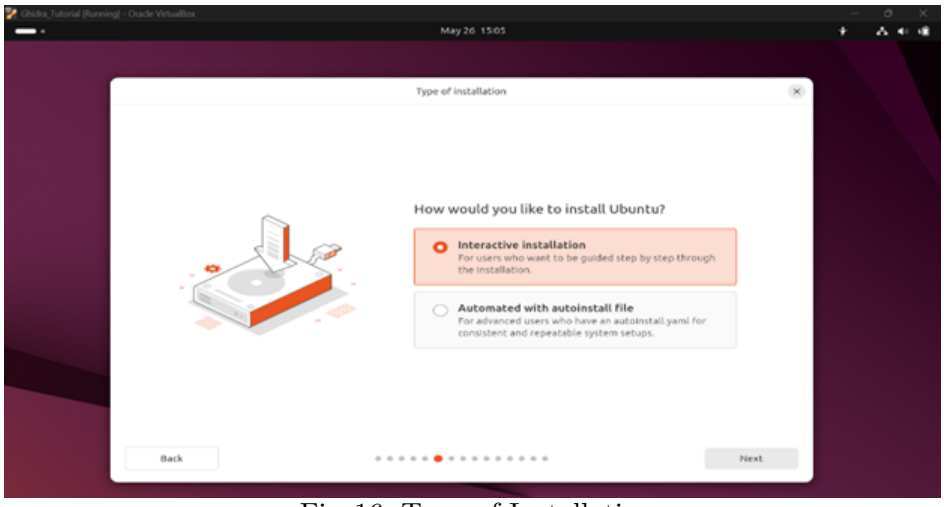

Fig. 16: Type of Installation

– Leave the default selection for software packages and click **Next**.

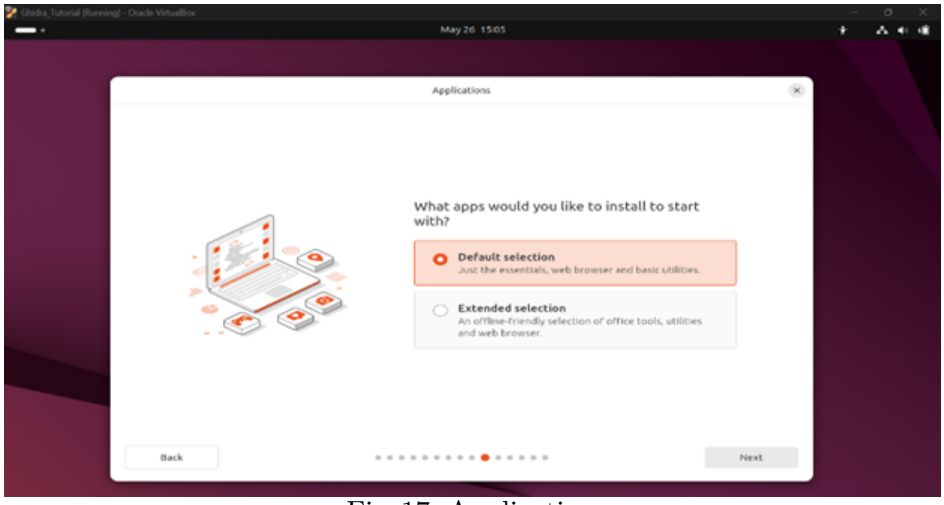

Fig. 17: Applications

– Check both boxes under **Install Recommended Proprietary Software** to ensure drivers and codecs are installed.

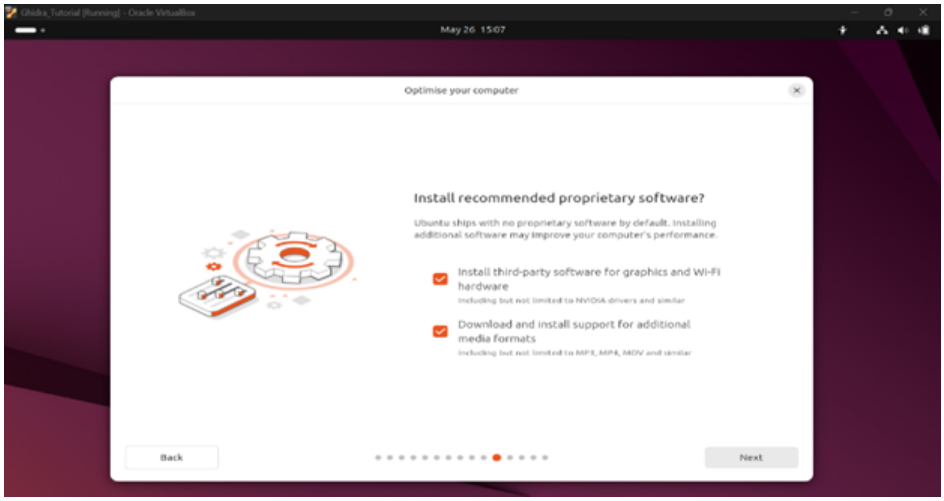

Fig. 18: Optimize your computer

– Choose **Erase disk and install Ubuntu** to perform a clean installation on the virtual drive.

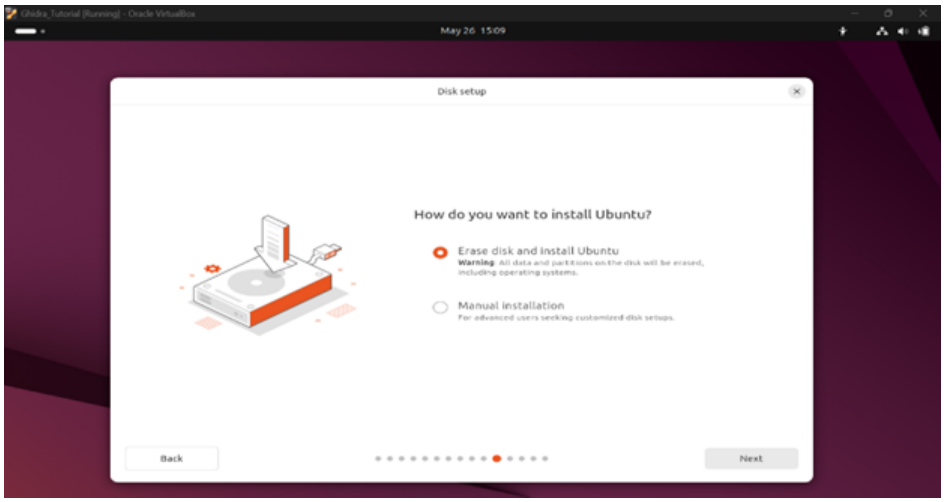

Fig. 19: Disk setup

– Select **No encryption** for the disk. Encryption is not necessary in this controlled environment.

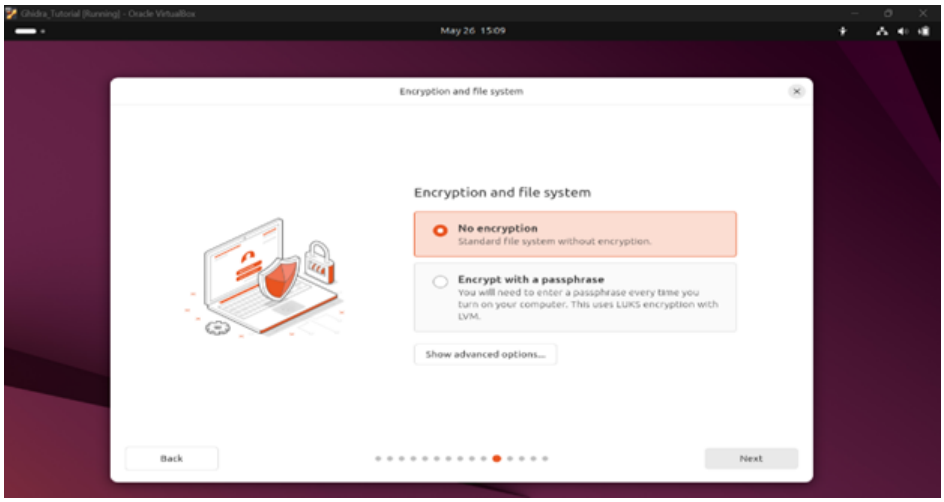

Fig. 20: Encryption and file system

– Under **Create your account**, enter a username and password you will remember (Or use the example in Fig.22). This account will be used to install Ghidra and other tools.

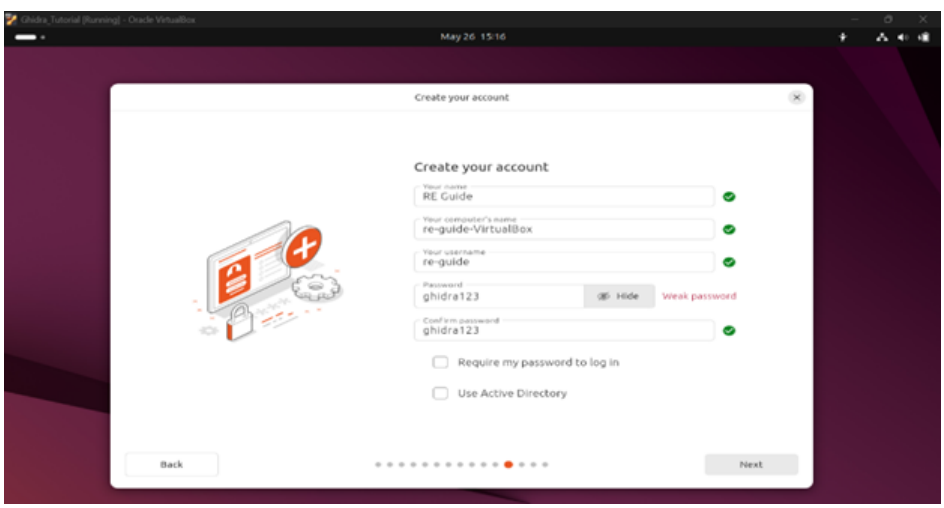

Fig. 21: Create your account

– Choose your timezone based on your location and click **Next**.

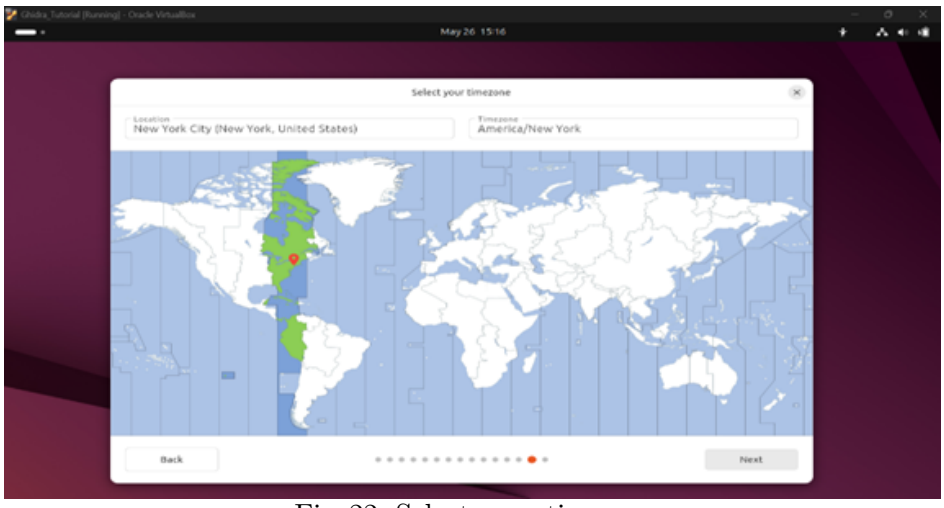

Fig. 22: Select your timezone

– Review your configuration settings and click **Install**. After installation is complete, reboot the virtual machine when prompted.

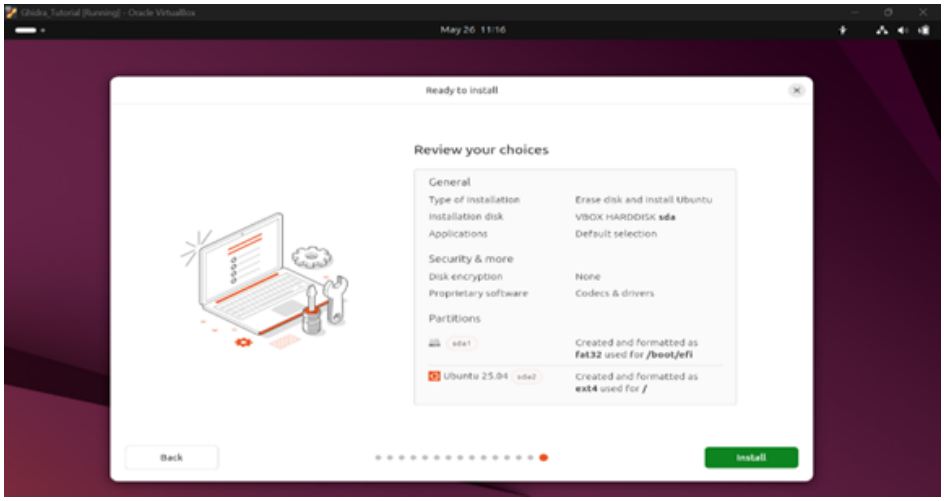

Fig. 23: Review your choices

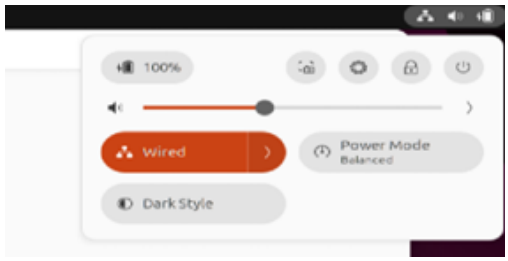

Fig. 24: Reboot Virtual Machine

4.2 Ghidra Installation

For the final step in this documentation, we will be installing **Ghidra**. Once the virtual machine has been rebooted we will open a new terminal window.

In the terminal, our first commands will be **sudo apt update** then **sudo apt upgrade**. This will ensure all the packages and libraries with in our virtual machine are updated and will not cause any problems in our installation process.

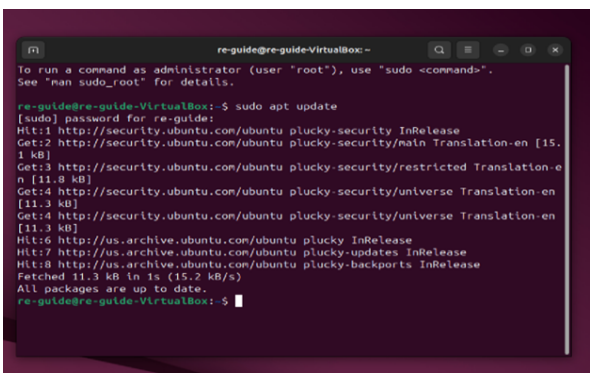

Fig. 25: **sudo apt update**

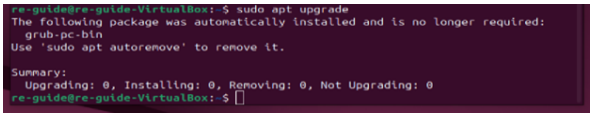

Fig. 26: **sudo apt upgrade**

Since Ghidra is a Java-based application, we will need to install a Java Development Kit (JDK) as a runtime. We will be installing the **openjdk** package from the official Ubuntu repository using the following command: **sudo apt install openjdk-21-jdk**.

To finish the installation, type **Y** when prompted and press **Enter**.

We will now download Ghidra. In the terminal, navigate to the **Downloads** directory using:

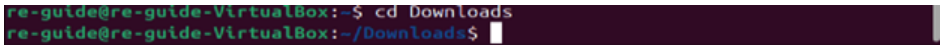

Fig. 28: **cd Downloads**

```
re-guide@re-guide-VirtualBox:~$ sudo apt install openjdk-21-jdk
The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'sudo apt autoremove' to remove it.

Installing:
  openjdk-21-jdk

Installing dependencies:
  ca-certificates-java      libx11-dev                openjdk-21-jre-headless
  fonts-dejavu-extra        libxau-dev                uuid-dev
  java-common               libxcb1-dev               x11proto-dev
  libatk-wrapper-java       libxdmcp-dev              xorg-sgml-doctools
  libatk-wrapper-java-jni   libxt-dev                 xtrans-dev
  libice-dev                openjdk-21-jdk-headless
  libsm-dev                 openjdk-21-jre

Suggested packages:
  default-jre  libxcb-doc          visualvm                | fonts-wqy-zenhei
  libice-doc   libxt-doc           fonts-ipafont-gothic    fonts-indic
  libsm-doc    openjdk-21-demo     fonts-ipafont-mincho
  libx11-doc   openjdk-21-source   fonts-wqy-microhei

Summary:
  Upgrading: 0, Installing: 20, Removing: 0, Not Upgrading: 0
  Download size: 137 MB
  Space needed: 318 MB / 3,204 MB available

Continue? [Y/n]
```

Fig. 27: **sudo apt install openjdk-21-jdk**

Then run the following command to download the latest version of Ghidra (11.3.2 as of May 2025):

```
wget https://github.com/NationalSecurityAgency/ghidra/releases/download/
Ghidra_11.3.2_build/ghidra_11.3.2_PUBLIC_20250415.zip
```

```
re-guide@re-guide-VirtualBox:~/Downloads$ sudo wget https://github.com/NationalS
ecurityAgency/ghidra/releases/download/Ghidra_11.3.2_build/ghidra_11.3.2_PUBLIC_
20250415.zip
```

Fig. 29: **wget command to download Ghidra**

After the download finishes, unzip the archive with the following command:
unzip ghidra_11.3.2_PUBLIC_20250415.zip

The extracted folder will be in the same directory. Navigate into it with:
cd ghidra_11.3.2_PUBLIC

Then run Ghidra using the included script: **./ghidraRun**

```
re-guide@re-guide-VirtualBox:~/Downloads$ cd ghidra_11.3.2_PUBLIC/
re-guide@re-guide-VirtualBox:~/Downloads/ghidra_11.3.2_PUBLIC$ ./ghidraRun
```

Fig. 30: **cd and ./ghidraRun command**

In a few seconds, Ghidra should prompt a user agreement window and run successfully.

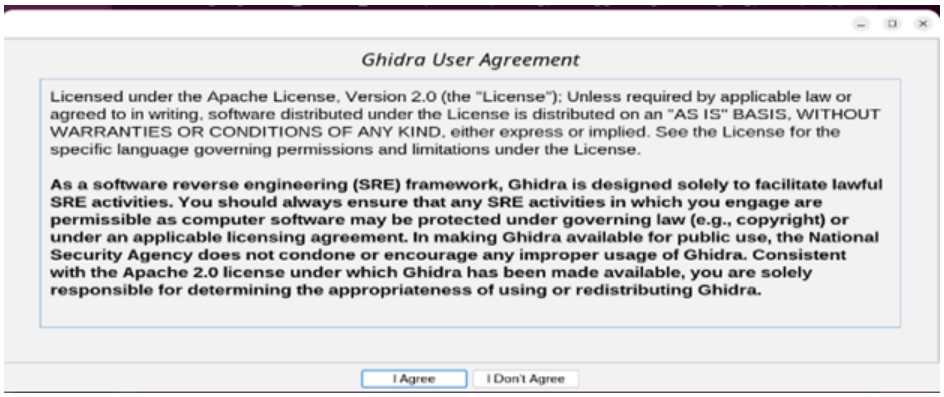

Fig. 31: **User Agreement Window**

Fig. 32: **Ghidra Loading Page**

This concludes Part 1 of out tutorial, the installation guide for Virtual Box, Ubuntu, and Ghidra. Please note that installing the OS might be tricky and a simple restart may fix any issues. Another note, when the virtual machine is not in use, it is best to shut it down for safety reasons and free allocated space to be use on the host computer. In the next document, we will dive into our first reverse engineering examples.